On a mixed GPU cluster, the parallelism plan decides how much of the hardware you use

Numbers from a published paper (Sailor, SOSP 2025). Concept, not client work.

Planners can misjudge memory badly before training even starts

95%

how far off existing planners can be on memory footprint

The Sailor authors found errors of 25% to 95% across planners, for OPT-350M on a Grace-Hopper cluster of identical GPUs.

Source: Strati et al., Sailor, SOSP 2025, arXiv:2504.17096, section 3.2

A plan built for identical GPUs misses what a mixed pool needs

PLAN FOR IDENTICAL GPUS	WHAT A MIXED POOL NEEDS
Every worker assumed the same speed	Speed and memory profiled per GPU type
Memory estimated from a formula	Stages placed by GPU type
Re-planned by hand when GPUs change	A new plan when availability changes

Sailor's plans beat other planners on mixed A100 and V100 GPUs

1.9x the throughput of AMP, simulated 50/50 A100 and V100, OPT-350M

1.15x the throughput of Metis on the same setup

1.08 to 1.81x over baselines on real GPUs, 8 per type

Source: Strati et al., Sailor, SOSP 2025, arXiv:2504.17096, section 5.2.2

Its estimates were close to measured, on mixed and on uniform clusters

4.5%

average iteration-time error on a mixed Titan-RTX, RTX-2080 and RTX-3090 cluster

5.56%

average memory error on a uniform GH200 cluster, against 12.5% to 74% for baselines

Source: Strati et al., Sailor, SOSP 2025, arXiv:2504.17096, section 5.1

How we would approach it: measure each GPU type, then plan

Illustrative, not a result

01 **Profile**

Measure throughput and memory per GPU type on your own model.

02 **Plan**

Size stages and batch shares from those profiles, not from datasheets.

03 **Check**

Compare real step time with the estimate, and re-plan when it drifts.

Our research collaboration also starts by measuring

> **Surveying how multi-GPU, multi-node work is split today, building a benchmark for mixed GPU setups, then proposing optimisations and measuring what each one actually changes.**

Our plan with a Norwegian research laboratory, in progress (hopline.co/work).

How do you plan parallelism across GPU types today?

Scoping is short, fixed and ends in a written plan.

contact@hopline.co